

Calculative Foundation Project Report

Vectors

- A vector represents a student's scores as an ordered list of numbers.
- Each student has 5 subject scores forming a 5-dimensional vector.
- Two vectors can be compared using dot product and angle between them.
- Cross product is used for 3D vectors to find a perpendicular direction.
- Projection shows how much one student vector aligns with another.

Examples:

Example 1: S001 vector = [61, 94, 75, 89, 56]

Example 2: Dot product of S001 and S002 = $61 \times 74 + 94 \times 53 + 75 \times 82 + 89 \times 49 + 56 \times 63$

Norms

- Norm gives the overall size or length of a score vector.
- L1 norm adds up all the absolute score values in the vector.
- L2 norm finds the straight-line distance from origin to the score point.
- Higher norm means the student has higher overall scores.
- Used to compare and normalize student score vectors.

Examples:

Example 1: L1 norm of [61, 94, 75, 89, 56] = 375

Example 2: L2 norm of [3, 4] = $\sqrt{9 + 16} = 5$

Matrix Operations

- Scores of multiple students form a matrix with rows as students and columns as subjects.
- Matrix addition combines scores of two groups of students element by element.
- Matrix multiplication computes similarity scores between groups of students.
- Transpose swaps rows and columns converting student-wise to subject-wise view.
- Inverse exists only when the determinant is non-zero.

Examples:

Example 1: Adding scores of student groups gives combined performance matrix

Example 2: Determinant non-zero confirms matrix is invertible

Linear Transformations and Geometry

- One subject score like Physics represents a 1D line.
- Two subjects like Physics and Chemistry form a 2D plane.
- Three subjects Physics, Chemistry and Biology create a 3D space.

- All 5 subjects together form a 5-dimensional hyperplane.
- Each subject added increases the number of dimensions by one.

Examples:

Example 1: Physics scores alone plotted on a number line is 1D

Example 2: Physics vs Chemistry scatter plot is a 2D plane

Eigenvalues and Eigenvectors

- Computed from the covariance matrix of the academic score data.
- Eigenvectors point in the directions of maximum data spread.
- Eigenvalues tell us how much variance exists in each direction.
- Subjects with higher eigenvalue contribution vary more across students.
- Form the mathematical basis for PCA dimensionality reduction.

Examples:

Example 1: Covariance matrix of 5 subjects gives 5 eigenvalues and eigenvectors

Example 2: Largest eigenvalue corresponds to the direction of greatest score variance

LU Decomposition

- Splits the score matrix into a lower triangular L and upper triangular U matrix.
- A permutation matrix P handles row reordering during decomposition.
- Makes solving linear systems faster and more numerically stable.
- Useful for repeated solving with the same matrix but different right-hand sides.
- Applied on the first 5 students score matrix in this project.

Examples:

Example 1: Score matrix $M = P \times L \times U$

Example 2: L has ones on diagonal, U has non-zero upper values

Singular Value Decomposition (SVD)

- Breaks the full score matrix into three matrices U, S and Vt.
- S contains singular values sorted from largest to smallest.
- Larger singular values capture more important patterns in the data.
- Keeping only top singular values reduces dimensions while keeping key info.
- Widely applied in image compression and recommendation engines.

Examples:

Example 1: Academic score matrix (200x5) gives 5 singular values

Example 2: First singular value is much larger than the rest showing one dominant pattern

PCA — Principal Component Analysis

- Reduces 5 subject scores to 2 principal components for visualization.
- New components are linear combinations of original subject scores.
- First component captures the direction of maximum score variance.
- Does not need class labels — works on raw score data directly.
- Helps identify clusters of High and Low performing students visually.

Examples:

Example 1: 200 students with 5 scores reduced to 200 students with 2 components

Example 2: PCA scatter plot separates High and Low performers into visible clusters

LDA — Linear Discriminant Analysis

- Supervised method that uses class labels High and Low to find best separation.
- Finds a single axis that maximizes distance between High and Low groups.
- Projects 5D score data onto 1D while keeping class separation intact.
- More effective than PCA for classification tasks with labeled data.
- Clearly separates High performance students from Low performance students.

Examples:

Example 1: Students scoring average above 70 labeled High, rest labeled Low

Example 2: LDA reduces 5 subject scores to one value that best splits both groups

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